Cognitive Reality Modeling Standard (CRMS)

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Cognitive Integrity Standard (CIS) ·
Cognitive Interpretation Integrity Standard (CIIS) ·
Cognitive Reasoning Integrity Standard (CRIS) ·
Agent Cognition Integrity Standard (ACIS) ·
Multi-Agent Cognition Integrity Standard (MCIS) ·
Multi-Layer Truth Validation Framework (MTVF) ·
Universal Canonical Language (UCL) ·
INTEGROS® — Integrity Standard
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Canonical Definition System

Canonical Definition

Cognitive Reality Modeling Standard (CRMS) defines the structural conditions under which cognition constructs, maintains, updates, validates, and governs internal representations of reality in a manner that remains traceable, coherent, reality-connected, adaptable, governance-valid, and operationally reliable across human, agent, and multi-agent cognition environments.

CRMS governs reality models.

The standard ensures that cognition remains capable of building usable representations of reality without allowing those representations to become detached from operational reality, corrupted by model drift, distorted by incomplete assumptions, or rendered governance-invalid through persistent divergence from observable conditions.

Reality is never accessed directly.

Reality is accessed through models.

CRMS governs the integrity of those models.

A. Standard Abstract

Every cognition system operates through reality models.

Humans use reality models.

Agents use reality models.

Organizations use reality models.

Societies use reality models.

Future autonomous systems will continuously build and update internal representations of:

- environments
- situations
- operational conditions
- future states
- risk conditions
- human behavior
- agent behavior
- reality itself

Most systems focus on:

- information
- interpretation
- reasoning
- decisions

However, beneath all of them sits a deeper layer:

The model of reality being used.

A system may:

- receive correct information
- interpret correctly
- reason correctly

and still fail because its internal reality model is wrong.

CRMS exists to govern that condition.

B. Core Principle

Cognition does not operate directly on reality. Cognition operates on models of reality.
Reality modeling becomes valid only when internal representations remain sufficiently aligned
with observable reality, traceable assumptions, evolving evidence,
and governance-valid adaptation conditions.

C. Scope

This standard may apply to:

- human cognition

- AI systems
- autonomous agents
- multi-agent systems
- digital twins
- simulation systems
- forecasting systems
- intelligence environments
- strategic planning systems
- future autonomous cognition ecosystems

CRMS applies wherever cognition constructs internal representations of reality.

D. Why This Standard Exists

Interpretation creates meaning.

Reasoning creates conclusions.

Reality models create the world within which reasoning occurs.

A cognition system may possess:

- valid information
- valid interpretation
- valid reasoning

while operating on a fundamentally flawed model of reality.

Traditional governance often evaluates outputs.

CRMS evaluates the model generating those outputs.

This distinction becomes critical as autonomous systems increasingly rely
on internal world models to navigate complexity.

CRMS exists because reality models themselves have become governable spaces.

E. Cognitive Reality Modeling Logic

Reality-model integrity exists only when the following remain materially preservable:

1. Reality Representation Integrity
Reality representations remain identifiable and traceable.

2. Model Alignment Integrity
Reality models remain sufficiently aligned with observable reality.

3. Model Adaptation Integrity
Reality models remain capable of legitimate adaptation.

4. Reality Drift Integrity
Model-reality divergence remains detectable and governable.

5. Predictive Reality Integrity
Future-state representations remain accountable to reality conditions.

F. Operational Architecture Space

CRMS defines the operational architecture space for:

- reality modeling governance
- world-model governance
- model adaptation governance
- model-reality alignment
- predictive reality modeling
- simulation-based cognition
- reality representation governance
- reality drift governance
- model validation

This space exists because cognition acts through models of reality rather than reality itself.

CRMS governs whether those models remain valid.

G. Difference Between Reasoning and Reality Modeling

Reasoning and reality modeling are related but structurally distinct.

Reasoning governs:

- inference
- logic
- evidence evaluation
- conclusion formation
- justification

Reality modeling governs:

- world representations
- environment representations
- future-state representations
- operational-reality representations
- model adaptation
- model-reality alignment

A cognition system may reason perfectly within an incorrect model.

CRMS governs the model itself.

H. Runtime Position

CRMS operates within the Cognitive Reality Modeling Governance Layer of CLIA®.

It governs the internal reality structures upon which interpretation, reasoning, planning, prediction, and decision-making depend.

CRMS therefore operates beneath many higher cognition functions.

I. Reality Drift Rule

Reality drift occurs when internal reality models progressively diverge from operational reality while cognition continues assuming the model remains valid.

This may include:

- outdated world models
- incorrect environment assumptions
- predictive model collapse
- simulation drift
- operational-reality divergence
- model rigidity
- hidden model errors
- reality-representation degradation

A cognition system may therefore remain internally coherent while progressively losing contact with reality.

CRMS exists to expose and govern that condition.

J. Validity Logic

A cognition environment is valid under CRMS when:

- reality models remain identifiable
- reality alignment remains assessable
- model assumptions remain visible
- adaptation remains possible
- drift remains detectable
- predictive models remain accountable
- reality representations remain governable

A cognition environment becomes invalid under CRMS when:

- models become detached from reality
- drift becomes invisible
- assumptions become unreconstructable
- adaptation collapses
- predictive validity deteriorates
- reality representations become ungovernable
- model legitimacy cannot be demonstrated

K. Relationship to Other OOF Standards

CRMS operates naturally with:

- Cognitive Integrity Standard (CIS)
- Cognitive Interpretation Integrity Standard (CIIS)
- Cognitive Reasoning Integrity Standard (CRIS)
- Agent Cognition Integrity Standard (ACIS)
- Multi-Agent Cognition Integrity Standard (MCIS)
- Multi-Layer Truth Validation Framework (MTVF)
- Universal Canonical Language (UCL)
- INTEGROS® — Integrity Standard

CRMS does not replace these standards.

It governs the reality-model layer through which cognition understands and navigates reality.

L. Foundational Principle

If reality models cannot remain aligned, adaptable, traceable, and accountable
to observable reality, cognition may preserve coherence
while progressively losing reality validity.

Canonical Closing Statement

Cognitive Reality Modeling Standard (CRMS) defines the structural conditions
under which cognition constructs, maintains, updates, validates, and governs internal
representations of reality in a manner that remains traceable, coherent, reality-connected,
adaptable, governance-valid, and operationally reliable across human, agent,
and multi-agent cognition environments.

Cognition does not operate directly on reality. Cognition operates on models of reality.
Reality modeling becomes governance-valid only when those models remain sufficiently aligned
with observable reality, evolving evidence, and accountable adaptation throughout cognition.

Modules

- RRIM→ Reality Representation Integrity Module
- MAIM→ Model Alignment Integrity Module
- MADIM→ Model Adaptation Integrity Module
- RDIM→ Reality Drift Integrity Module
- PRIM→ Predictive Reality Integrity Module

Related Documents

- → About the Cognitive Reality Modeling Standard (CRMS)
- → CLIA® Architecture Index
- → CLIA® Complete Standards & Modules Index

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